*"Same code, every environment -- configuration should change, never the codebase."*

MODULE 26

# Spring Configuration, Profiles and Environments

Manage Spring Boot application configuration effectively across development, test, and production using profiles, externalized properties, and secure secrets handling.

TEKsystems Global Services

# WORKFORCE DEVELOPMENT

# MODULE 26 OVERVIEW

*Manage Spring Boot application configuration effectively across development, test, and production using profiles, externalized properties, and secure secrets handling.*

Configuration -- not code -- should change between environments. This module covers YAML/properties, Spring Profiles, property-source precedence, and secrets handling, then locks down Northstar CRM's dev/test/prod environments.

## DURATION & LAB

### 3 Hours Theory

application.yml vs application.properties, Spring Profiles, property-source precedence, and @ConfigurationProperties.

### 90 Minutes Lab

Northstar CRM Environments: build dev/test/prod profile files with environment-variable secrets.

### Scenario

Dev settings leaking into production (open H2 console, blank DB password) are unacceptable -- prod must fail fast.

## MODULE ARC

### Understand Precedence

CLI args beat environment variables beat profile YAML beat application.yml beat code defaults.

### Master Profiles & Secrets

Activate profiles two ways; bind @ConfigurationProperties; never hard-code secrets.

### Build the Lab

Convert Northstar CRM to application.yml with dev/test/prod profiles and environment-variable credentials.

**KEY TAKEAWAY** Total time: 3 hours theory + 90 minutes hands-on lab. Good configuration leads to flexible applications; secure configuration leads to trustworthy ones.

# WHY ENVIRONMENT MANAGEMENT MATTERS

*Applications run in different environments -- development, testing, staging, and production -- each with unique requirements and constraints.*

Effective environment management ensures your application is flexible, secure, and reliable across the entire delivery lifecycle.

## THE CHALLENGE WITHOUT PROPER ENVIRONMENT MANAGEMENT

- **Hard-coded Values**
- **Security Risks**
- **Inconsistent Behavior**
- **Deployment Delays**
- **Higher Operational Cost**

## THE BENEFITS OF EFFECTIVE ENVIRONMENT MANAGEMENT

### Consistency

Maintain consistent behavior across all environments; reduce environment-specific issues.

### Security

Keep sensitive data out of source code with secure, externalized configuration.

### Flexibility

Switch configurations across environments without any code changes.

### Faster Delivery

Simplify builds and promote code through environments with confidence.

### Maintainability

Centralize and organize configuration; easier to manage as applications grow.

### Team Productivity

Developers focus on features, not environment setup; fewer errors, more value.

**KEY TAKEAWAY** Environment management is not just about configuration -- it's about building applications that are portable, secure, and enterprise-ready.

# MODULE LEARNING OBJECTIVES

*By the end of this module, you will be able to effectively manage Spring Boot application configuration across multiple environments using profiles and externalized properties.*

- **1. Understand Configuration Fundamentals —** explain the role of configuration in Spring Boot applications and compare application.properties and application.yml.
- **2. Work with YAML Configuration —** use YAML syntax and structure to organize and manage application configuration effectively.
- **3. Use Spring Profiles —** create and activate profiles for development, test, and production environments.
- **4. Manage Environment Configuration —** leverage environment variables, command-line arguments, and external property sources to externalize configuration.
- **5. Secure Sensitive Information —** apply best practices to protect sensitive data and explore basic secrets management strategies.
- **6. Apply Enterprise Configuration Practices —** implement configuration strategies across environments and follow best practices for maintainable, secure deployments.

**KEY TAKEAWAY** Master Spring Profiles and configuration to build flexible, secure, and environment-aware applications.

# CONFIGURATION MANAGEMENT OVERVIEW (1 OF 2)

*Configuration management is the practice of managing application settings and externalizing them from code so applications can adapt to different environments.*

## WHY IT MATTERS

### Flexibility

Change settings without modifying or rebuilding code.

### Reliability

Ensure consistent behavior across environments.

### Scalability

Support multiple environments and deployments.

### Security

Protect sensitive information using secure practices.

### Cost Efficiency

Reduce errors and downtime from misconfiguration.

## KEY PRINCIPLES

- **Externalize Configuration —** store configuration outside the application so it can be changed depending on the environment.
- **Environment Awareness —** use environment-specific settings for development, test, staging, and production.
- **Separation of Concerns —** keep configuration separate from business logic and source code.
- **Centralized and Consistent —** manage configuration in a consistent and organized way across services and teams.
- **Secure and Governed —** protect sensitive data and follow governance for configuration changes.

**KEY TAKEAWAY** Configuration management is the practice of managing application settings and externalizing them from code so applications can adapt to different environments.

# Configuration Management Overview

Spring Boot provides a powerful and flexible way to externalize configuration, manage different environments using profiles, and resolve properties in a defined order of precedence.

## 1. CONFIGURATION SOURCES

- **application.properties / application.yml**
  Default configuration
- **application-{profile}.properties / application-{profile}.yml**
  Profile-specific configuration
- **External Config Files**
  File system, classpath, or config server
- **Environment Variables**
  System environment variables
- **Command Line Arguments**
  --key=value
- **Cloud / Secrets / Vault**
  AWS Parameter Store, Vault, etc.

Multiple sources. One unified model.

## 2. SPRING ENVIRONMENT

Spring Boot

- **Configurable Environment**
  Abstraction over property sources
- **Active Profiles**
  dev, test, prod, ...
- **Property Sources**
  Ordered collection of sources

## 3. PROPERTY RESOLUTION

Precedence Order

HIGHEST

1. Command Line Arguments
2. Environment Variables
3. External Config Files
4. Profile-specific Files
5. application.properties / application.yml (default)
6. Spring Boot Defaults

LOWEST

Higher priority sources override lower priority sources.

## 4. USED BY APPLICATION

- **@Value Injection**
  @Value("${key}")
- **@ConfigurationProperties**
  Type-safe configuration
- **Auto-configuration**
  Uses properties internally
- **Beans & Components**
  Configured behavior

Configuration drives behavior. No code changes between environments.

- **Externalized Configuration**
  Keep config separate from code
- **Environment Flexibility**
  Different settings for different environments
- **Centralized Management**
  Consistent and maintainable configuration
- **Secure Configuration**
  Protect sensitive information
- **Cloud & Enterprise Ready**
  Integrates with modern infrastructure

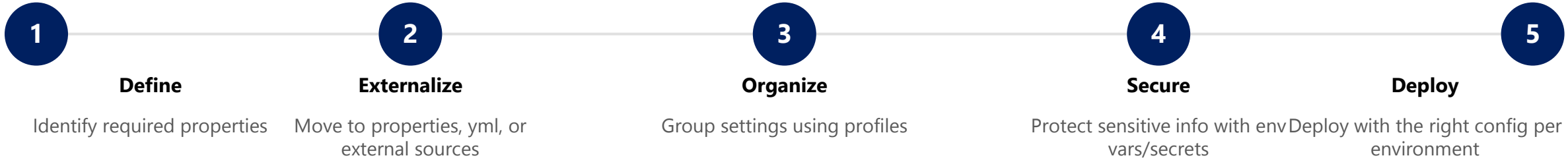
**Goal**

Right configuration, in the right place, for the right environment.

# CONFIGURATION MANAGEMENT OVERVIEW (2 OF 2)

*The configuration management flow, and the typical sources configuration comes from in a Spring Boot application.*

## CONFIGURATION MANAGEMENT FLOW




## TYPICAL CONFIGURATION SOURCES

| Source | Description |
|---|---|
| **application.properties / application.yml** | Default configuration files packaged with the application. |
| **Environment Variables** | Environment-specific values provided by the OS or container platform. |
| **Command-line Arguments** | Override properties at startup using --key=value. |
| **External Configuration Servers** | Centralize configuration using systems like Spring Cloud Config. |
| **Secrets Management Systems** | Store and retrieve sensitive information securely (e.g. AWS Secrets Manager). |

**KEY TAKEAWAY** Goal: deliver the right configuration to the right environment at the right time.

# APPLICATION.PROPERTIES

*The default Spring Boot configuration file, using a simple key-value format.*

**Example: application.properties**

```
# Application identity
spring.application.name=DemoApp

# Server configuration
server.port=8080
server.servlet.context-path=/api

# Database configuration
spring.datasource.url=jdbc:mysql://localhost:3306/demo
spring.datasource.username=dbuser
spring.datasource.password=dbpassword

# JPA / Hibernate
spring.jpa.hibernate.ddl-auto=update

# Logging
logging.level.root=INFO
```

## KEY CHARACTERISTICS

- **Simple Key-Value Format —** uses dot notation for hierarchical properties.
- **Auto Loaded by Spring Boot —** automatically detected from src/main/resources.
- **Supports Profiles —** override properties per environment (application-dev.properties).
- **Externalizable —** can be overridden by environment variables or command-line args.

## COMMON USE CASES

**Database Connections** **Server/Port Settings**

**Logging Levels** **Feature Flags**

**KEY TAKEAWAY** Use application.properties for simplicity; use application.yml when you need more complex or structured configuration.

# APPLICATION.YML

*A YAML-based alternative for cleaner, hierarchical configuration.*

**Example: application.yml**

```yaml
spring:
  application:
    name: DemoApp
server:
  port: 8080
  servlet:
    context-path: /api
datasource:
  url: jdbc:mysql://localhost:3306/demo
  username: dbuser
  password: dbpassword
jpa:
  hibernate:
    ddl-auto: update
logging:
  level:
    root: INFO
```

## YAML vs PROPERTIES

| application.properties | application.yml |
|---|---|
| **Flat key-value format** | Hierarchical structure |
| **Harder to read at scale** | More readable |
| **Repetitive keys** | Less repetition |

## KEY BENEFITS

- **Hierarchical Structure —** natural nesting of related properties.
- **Less Duplication —** avoids repeating long property name prefixes.
- **Profile Support —** easy to define environment-specific configuration.

**KEY TAKEAWAY** Spring Boot automatically detects and loads application.yml if present in src/main/resources, giving it the same priority as application.properties.

# YAML SYNTAX FUNDAMENTALS

*YAML is a human-friendly data serialization format that uses indentation and structure instead of brackets and braces.*

## CORE SYNTAX RULES

- **Indentation Matters —** YAML uses spaces (never tabs) to define structure and nesting.
- **Key: Value Pairs —** keys and values are separated by a colon and a space.
- **Lists —** defined using a hyphen (-) followed by a space.
- **Strings —** can be plain, single-quoted, or double-quoted.
- **Comments —** use # to add a comment; everything after # is ignored.

**Example: application.yml**

```yaml
spring:
  application:
    name: DemoApp
server:
  port: 8080
datasource:
  url: jdbc:mysql://host/db
features:
  - payments
  - notifications
logging:
  level:
    root: INFO
    com.example: DEBUG
```

## DATA TYPES IN YAML

| Type | Example |
|---|---|
| **String** | name: DemoApp |
| **Number** | port: 8080 |
| **Boolean** | enabled: true |
| **Null** | host: null |
| **List** | - payments |
| **Map** | database: {...} |

## BEST PRACTICES

**2-space Indentation** | **Avoid Tabs** | **Meaningful Key Names** | **Quote Strings Only When Needed**

**KEY TAKEAWAY** YAML is case-sensitive. Correct indentation ensures your configuration is loaded properly.

# WHAT ARE SPRING PROFILES? (1 OF 2)

*Spring Profiles allow you to define different configurations for different environments and activate the appropriate configuration at runtime.*

Profiles help you build flexible applications that can adapt to environment-specific settings without changing the code.

## HOW IT WORKS

1. **1. Define Profiles**
   Create config files per environment
2. **2. Activate Profile**
   Specify the active profile
3. **3. Load Configuration**
   Spring loads the matching config
4. **4. Run Application**
   App runs with those settings applied

## KEY BENEFITS

- Environment-specific configuration with no code changes to switch environments.
- Cleaner, more maintainable codebase, and easier deployment across environments.
- Supports dev, test, staging, prod, and more.

**KEY TAKEAWAY** Spring Profiles let you switch entire configurations at runtime, without touching a single line of code.

# Spring Profiles and Environment Flow

Manage different environments (dev, test, stage, prod) with Spring Profiles

### 1 Application Startup

Spring Boot application starts.

Active profile is determined from configuration, system properties, or environment variables.

### 2 Profile Resolution

Spring resolves the active profile(s).

Default profile is used if no active profile is set (default: "default").

### 3 Property Source Loading

Spring loads property sources in order of precedence.

Profile-specific property files are loaded (e.g., application-dev.yml).

### 4 Bean Definition Filtering

Beans and configuration classes are included or excluded based on @Profile conditions.

Only beans matching the active profile(s) are registered.

### 5 ApplicationContext Intiialized

ApplicationContext is created with environment-specific beans and properties.

Ready to serve requests.

### 6 Environment Ready

Application runs with the configuration, beans, and resources specific to the active environment.

## HOW PROFILES ARE ACTIVATED

**Command Line**
`--spring.profiles.active=dev`

**System Property**
`-Dspring.profiles.active=dev`

**Environment Variable**
`SPRING_PROFILES_ACTIVE=dev`

**application.properties / .yml**
`spring.profiles.active=dev`

## PROFILE-SPECIFIC CONFIGURATION FILES

**Base Configuration (Always Loaded)**

application.yml

```
app:
  name: MyApp
logging:
  level:
    root: INFO
```

**Profile Specific (Loaded when active)**

application-dev.yml

```
server:
  port: 8080
datasource:
  url: jdbc:h2:mem:devdb
logging:
  level:
    root: DEBUG
```

application-test.yml

```
server:
  port: 8081
datasource:
  url: jdbc:h2:mem:testdb
logging:
  level:
    root: INFO
```

application-stage.yml

```
server:
  port: 8082
datasource:
  url: jdbc:postgresql://
    stage-db/mydb
logging:
  level:
    root: WARN
```

application-prod.yml

```
server:
  port: 80
datasource:
  url: jdbc:postgresql://
    prod-db/mydb
logging:
  level:
    root: ERROR
```

**Active Profile Example**

Active Profile: dev

**Effective Configuration**

```
server.port=8080
app.name=MyApp
datasource.url=jdbc:h2:mem:devdb
logging.level.root=DEBUG
...
```

## @Profile USAGE

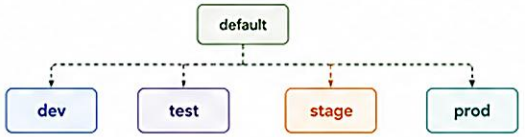
```
@Profile("dev")
@Configuration
public class DevConfig {
  // dev specific beans
}

@Profile({"stage","prod"})
@Service
public class AuditService {
  // used in stage and prod
```

## DEFAULT PROFILE

If no active profile is set, Spring uses the default profile: "default".

## PROFILE HIERARCHY



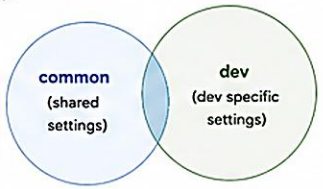

Properties from specific profile override those from the default profile.
Custom hierarchies can be defined using spring.profiles.include.

## MULTIPLE ACTIVE PROFILES

You can activate multiple profiles (comma separated).
Order of precedence (last one wins for conflicts).

**Example**

```
--spring.profiles.active=common,dev
# dev properties override common if same key exists
```




## KEY BENEFITS

**Environment Isolation**
Keep configurations separate for each environment.

**Easy Configuration Management**
Change settings without changing code.

**Safe Deployments**
Different settings for dev, test, stage, and prod.

**Scalability**
Supports complex applications with multiple environments.

**Best Practice:** Use profiles to externalize environment-specific configuration and keep your application portable and maintainable.

spring by VMware

# Spring Profiles

Spring Profiles let you define and activate different sets of configuration for different environments.

## What are Spring Profiles?

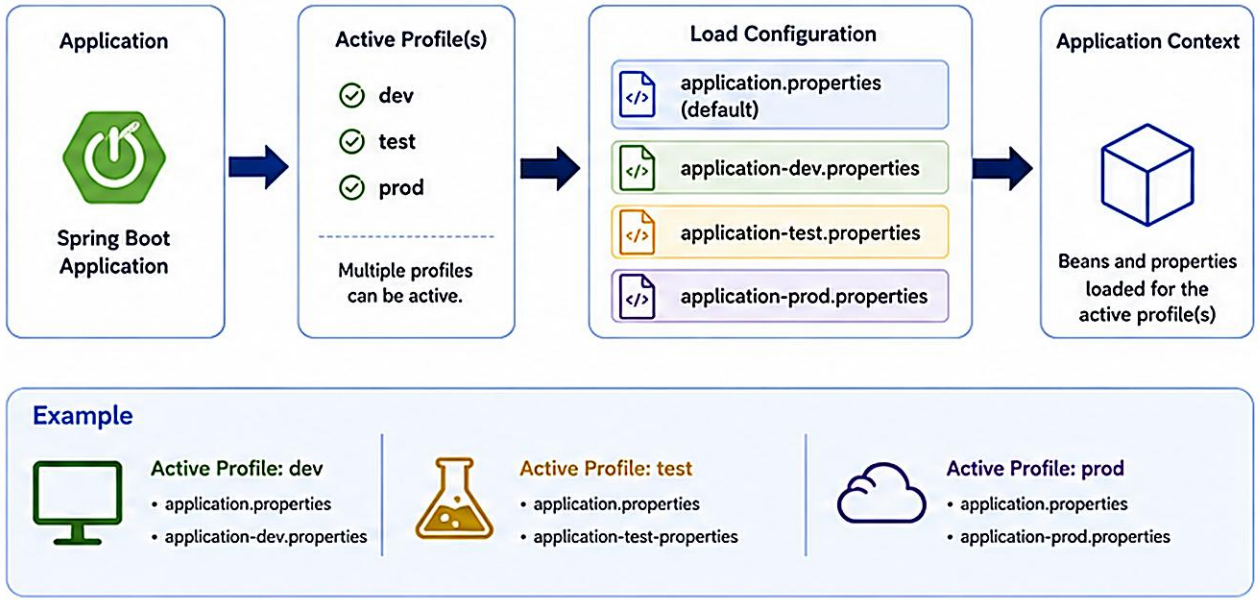
Profiles allow you to isolate environment-specific configuration.

Only the active profile's properties and beans are loaded.

Helps manage different environments using the same codebase.

**Key Benefit**
Cleaner configuration, safer deployments, and easier environment management.

## How Profiles Work




### Example

**Active Profile: dev**
- application.properties
- application-dev.properties

**Active Profile: test**
- application.properties
- application-test-properties

**Active Profile: prod**
- application.properties
- application-prod.properties

## Built-in & Custom Profiles

### Built-in Profiles

**default**
Used when no other profile is active.

**test**
Convention for test environment.

**prod**
Convention for production environment.

### Custom Profiles

Create any profile that fits your needs.
Examples: dev, qa, staging, uat, local

## Profile Activation

### 1. application.properties / application.yml

```
spring.profiles.active=dev
# or in YAML
spring:
  profiles:
    active: dev
```

OR

### 2. Command Line Argument

```
--spring.profiles.active=dev
java -jar app.jar \
  --spring.profiles.active=dev
```

OR

### 3. Environment Variable

```
SPRING_PROFILES_ACTIVE=dev
# Linux / Mac
export SPRING_PROFILES_ACTIVE=dev
# Windows (Command Prompt)
set SPRING_PROFILES_ACTIVE=dev
```

OR

### 4. Programmatically

```
SpringApplication app =
  new SpringApplication(App.class);
app.setAdditionalProfiles("dev");
app.run(args);
```

**Tip:** You can activate multiple profiles by comma-separated values (e.g., dev,common) for shared configuration.

**Note:** Profile-specific files override the default application.properties.

# WHAT ARE SPRING PROFILES? (2 OF 2)

*Profile configuration sources in order of precedence, profile-specific file naming, and how to activate a profile.*

## PROFILE CONFIGURATION SOURCES (IN ORDER OF PRECEDENCE)

- 1. Command-line arguments
- 2. Java System properties
- 3. OS environment variables
- 4. application.properties / application.yml
- 5. Default profile (when no profile is active)

### Example Activation

```
--spring.profiles.active=dev
-Dspring.profiles.active=test
SPRING_PROFILES_ACTIVE=prod

# Default profile name: default
```

## EXAMPLE: PROFILE-SPECIFIC FILES

- **application.yml —** common configuration shared by every environment.
- **application-dev.yml —** development-specific settings.
- **application-test.yml —** test-specific settings.
- **application-prod.yml —** production-specific settings.

## IMPORTANT

Only one profile is normally active at a time (profiles can be combined via profile groups). If no profile is specified, Spring uses the default profile.

**KEY TAKEAWAY** src/main/resources/application-{profile}.yml is automatically loaded by Spring Boot when that profile is active.

# DEVELOPMENT PROFILE

*Used during local development -- debug logging, in-memory databases, relaxed security.*

**Example: application-dev.yml**

```yaml
spring:
  profiles:
    active: dev
  datasource:
    url: jdbc:h2:mem:devdb;DB_CLOSE_DELAY=-1
    driver-class-name: org.h2.Driver
  h2:
    console:
      enabled: true
  jpa:
    hibernate:
      ddl-auto: update
    show-sql: true
logging:
  level:
    root: DEBUG
```

## KEY CHARACTERISTICS

- Optimized for local development and fast feedback.
- Uses in-memory databases (e.g. H2) and mock services.
- Enables detailed logging and debugging.
- Relaxed security for easier development.
- May include sample data and developer tools.

## HOW TO ACTIVATE

`-Dspring.profiles.active=dev`

`SPRING_PROFILES_ACTIVE=dev`

**KEY TAKEAWAY** Never use the dev profile in production. It is meant only for local developer environments.

# Spring Profiles: Development, Test, Production

Use Spring Profiles to keep configuration, beans, and resources environment-specific and promote safely from development to production.

## 1. DEVELOPMENT (dev)

For local development and feature work

### Purpose

Fast iteration, local tools, debug enabled

### Typical Configuration

- Debug logging (DEBUG/TRACE)
- H2 / Local DB
- Mock / Stub external services
- Relaxed security
- Dev-only features enabled

### Example Properties

```
spring.profiles.active=dev
server.port=8080
logging.level.root=DEBUG
spring.datasource.url=jdbc:h2:mem:devdb
feature.beta.enabled=true
```

### Common Tools

IntelliJ / VS Code · H2 DB · Docker (Local) · Spring DevTools

## 2. TEST (test)

For automated and manual testing

### Purpose

Validate functionality, integrations and quality

### Typical Configuration

- Info logging
- Test database (isolated)
- Real or mock integrations
- Test security setup
- Test data and seed scripts

### Example Properties

```
spring.profiles.active=test
server.port=8081
logging.level.root=INFO
spring.datasource.url=jdbc:postgresql://
   test-db:5432/app_test
feature.beta.enabled=false
```

### Common Tools

Selenium Tests · Docker (Test) · Test DB · CI Pipeline

## 3. PRODUCTION (prod)

For live users and business operations

### Purpose

Stability, security, performance and reliability

### Typical Configuration

- Warn / Error logging
- Production database
- Real external services
- Strict security
- Performance optimizations

### Example Properties

```
spring.profiles.active=prod
server.port=80
logging.level.root=WARN
spring.datasource.url=jdbc:postgresql://
   prod-db:5432/app
feature.beta.enabled=false
```

### Common Tools

AWS / Cloud · Kubernetes · Monitoring (Prometheus) · Secrets Manager

## Environment Promotion Flow

Develop (dev) → Build & Test (test) → Deploy (prod) → Monitor & Optimize

## Best Practices

- Keep configs externalized and environment-specific.
- Never store secrets in code or configuration files.
- Promote through the same pipeline.
- Use the same codebase across all environments.
- Monitor, log and alert in all environments.
- Document configuration changes and versions.

**Key Takeaway:** Profiles let you adapt the same application to different environments with the right configuration, improving reliability and reducing risk.

# TEST PROFILE

*Used for automated testing -- a stable, isolated environment for fast, repeatable tests.*

**Example: application-test.yml**

```yaml
spring:
  profiles:
    active: test
  datasource:
    url: jdbc:h2:mem:testdb;DB_CLOSE_DELAY=-1
    driver-class-name: org.h2.Driver
  h2:
    console:
      enabled: false
  jpa:
    hibernate:
      ddl-auto: create-drop
    show-sql: false
logging:
  level:
    root: WARN
```

## KEY CHARACTERISTICS

- Uses in-memory databases (e.g. H2) created fresh for each test run.
- Loads test data and mocks external dependencies.
- Optimized for automated tests -- speed and repeatability.
- Reduced security complexity for testing ease.
- Ensures consistent test execution across environments.

## HOW TO ACTIVATE

**-Dspring.profiles.active=test**

**SPRING_PROFILES_ACTIVE=test**

**KEY TAKEAWAY** Never use the test profile in staging or production. It is designed only for testing environments.

# Test Profile

Isolated, Fast, and Reliable Testing with Spring Profiles

The **Test Profile** provides a dedicated configuration for running tests in isolation, ensuring fast execution, predictable results, and no impact on other environments.

### Purpose

To define a separate configuration for the testing environment so that tests run against in-memory databases, mocked services, and safe test data.

## 1. TEST PROFILE CONFIGURATION

### Activate Test Profile

```
@ActiveProfiles("test")
@SpringBootTest
public class CustomerServiceTest {
    // test cases
}
```

### application-test.yml

```
spring:
  datasource:
    url: jdbc:h2:mem:testdb;DB_CLOSE_DELAY=-1
    driver-class-name: org.h2.Driver
    username: sa
    password:
  jpa:
    hibernate:
      ddl-auto: create-drop
      show-sql: true
server:
  port: 0
logging:
  level:
    root: INFO
```

### Profile Activation Sources

- @ActiveProfiles("test")
- spring.profiles.active=test
- --spring.profiles.active=test
- Maven/Gradle test configuration

## 2. HOW TEST PROFILE WORKS

1. **Activate** — Test starts with active profile "test"
2. **Load Config** — Spring loads application-test.yml and profile-specific beans
3. **Create Context** — Application context is initialized with test configuration
4. **Run Tests** — Tests execute using in-memory DB, mocked services, and test data
5. **Tear Down** — Context is closed and in-memory resources are released

## 3. WHAT THE TEST PROFILE PROVIDES

| In-Memory Database | Mocked Services | Ephemeral Server | Test Data | Test Logging |
|---|---|---|---|---|
| Uses H2 or HSQLDB for fast, isolated database operations. | External services are mocked using @MockBean or test doubles. | Server runs on random port (port: 0) to avoid port conflicts. | Uses predefined, safe, and repeatable test datasets. | Appropriate log levels to keep test output clean and readable. |

## 4. BENEFITS

- **Fast** — In-memory DB and mocked services reduce test execution time.
- **Isolated** — Tests run in isolation without affecting development or production data.
- **Reliable** — Consistent and repeatable test results every time.
- **Safe** — No risk of modifying real data or calling real external services.
- **CI/CD Friendly** — Optimized for automated test pipelines and parallel execution.

## 5. PROFILE HIERARCHY (PRECEDENCE)

Default < test < local < dev < prod

Higher precedence profiles override properties from lower precedence profiles.

## 6. EXAMPLE – TEST-SPECIFIC BEANS

### Test Configuration Class

```
@TestConfiguration
public class TestConfig {
    @Bean
    public EmailService emailService() {
        return new MockEmailService();
    }
}
```

### Mock Bean in Test

```
@SpringBootTest
@ActiveProfiles("test")
public class OrderServiceTest {
    @MockBean
    private PaymentGateway paymentGateway;
}
```

## 7. BEST PRACTICES

- Use separate application-test.yml for test configuration.
- Use in-memory databases (H2, HSQLDB) for tests.
- Mock external services using @MockBean.
- Use @ActiveProfiles("test") in test classes.
- Keep test data minimal and deterministic.
- Clean up data between tests (use @Transactional or @DirtiesContext when needed).
- Ensure tests are independent and can run in any order.

# PRODUCTION PROFILE

*Used in the live environment -- optimized for performance, security, and reliability.*

**Example: application-prod.yml**

```yaml
spring:
  profiles:
    active: prod
  datasource:
    url: jdbc:postgresql://prod-db.internal:5432/bankdb
    username: bankapp
    password: ${DB_PASSWORD}
    hikari:
      maximum-pool-size: 20
  jpa:
    hibernate:
      ddl-auto: validate
    show-sql: false
logging:
  level:
    root: WARN
```

## KEY CHARACTERISTICS

- Uses real, highly available databases and external systems.
- Hardened security settings and sensitive data management.
- Performance optimized with connection pooling, caching, and tuning.
- Minimal logging (INFO/WARN/ERROR) for performance.
- Designed for stability, scalability, and high availability.

## HOW TO ACTIVATE

**-Dspring.profiles.active=prod**

**SPRING_PROFILES_ACTIVE=prod**

**KEY TAKEAWAY** The production profile should be locked down, monitored, and thoroughly tested before deployment.

# Production Profile

Secure, Scalable, and Reliable Configuration for Live Environments

The **Production Profile** provides optimized and secure configuration for the live environment, ensuring performance, availability, monitoring, and data protection.

**Purpose**

To define a dedicated configuration for the production environment that ensures security, stability, scalability, observability, and reliability for real users and real data.

## 1. PRODUCTION PROFILE CONFIGURATION

### Activate Production Profile

```
@SpringBootApplication
@Profile("prod")
public class Application {
  public static void main(String[] args) {
    SpringApplication.run(Application.class, args);
  }
}
```

### application-prod.yml

```
spring:
  datasource:
    url: jdbc:postgresql://prod-db.company.com:5432/appdb
    username: ${DB_USERNAME}
    passswrd: ${DB_PASSWORD}
    hikari:
      maximum-pool-size: 20
      minimum-idle: 5
jpa:
  hibernate:
    ddl-auto: validate
    show-sql: false
server:
  port: 8080
logging:
  level:
    root: WARN
    com.company: INFO
```

### Profile Activation Sources

- @ActiveProfiles("prod")
- spring.profiles.active=prod
- --spring.profiles.active=prod
- OS Environment Variable: SPRING_PROFILES_ACTIVE=prod
- Docker / Kubernates environment variables

## 5. PROFILE HIERARCHY (PRECEDENCE)

default < prod < local < dev < test

Higher precedence profiles override properties from lower precedence profiles.

## 2. HOW PRODUCTION PROFILE WORKS

1. **Activate** — Application starts with active profile "prod"
2. **Load Config** — Spring loads application-prod.yml and profile-specific beans
3. **Create Context** — Application context is initialized with production configuration
4. **Run Application** — Application runs with optimized settings, real services, and external resources
5. **Monitor & Operate** — Application is monitored, logged, and scaled in production

## 3. WHAT THE PRODUCTION PROFILE PROVIDES

**External Databases**
Uses production-grade databases (PostgreSQL, Oracle, MySQL, etc.).

**Externalized Config**
All sensitive properties managed via environment variables or secrets (Vault, AWS Secrets).

**Security Hardened**
Enforces HTTPS, strong passwords, secure headers, and access controls.

**Optimized Resources**
Connection pooling, caching, threading, and memory tuned for load.

**Monitoring & Logging**
Structured logging, metrics, tracing, and health checks enabled.

## 6. EXAMPLE – PRODUCTION BEAN CONFIGURATION

### Production DataSource Bean

```
@Configuration
@Profile("prod")
public class DataSourceConfig {
    @Bean
    @ConfigurationProperties(prefix = "spring.datasource")
    public DataSource dataSource() {
        return DataSourceBuilder.create().build();
    }
}
```

### Production Properties via Environment Variables

```
# Environment Variables
DB_URL=jdbc:postgresql://prod-db.company.com:5432/appdb
DB_USERNAME=appuser
DB_PASSWORD=********
SPRING_PROFILES_ACTIVE=prod
SERVER_PORT=8080
LOG_LEVEL_ROOT=WARN
```

## 4. BENEFITS

**Secure**
Sensitive data is externalized and protected. HTTPS, strong headers, and secure defaults.

**Reliable**
Optimized connection pools, timeouts, and health checks ensure high availability.

**High Performance**
Tuned resources, caching, and efficient logging improve performance.

**Observable**
Health endpoints, metrics, and centralized logging enable full visibility.

**Scalable**
Designed to scale with load using external services and stateless configuration.

## 7. BEST PRACTICES

- Use application-prod.yml for all production-specific settings.
- Externalize all sensitive information.
- Never commit secrets to source control.
- Use connection pooling and set proper timeouts.
- Set logging to WARN or ERROR in production.
- Enable health, metrics, and distributed tracing.
- Test configuration in a staging environment first.
- Regularly rotate credentials and secrets.
- Backup, monitor, and review configurations continuously.

## KEY TAKEAWAYS

- Production profile isolates live configuration from all other environments.
- Keep configuration secure, externalized, and environment-specific.
- Optimize for performance, availability, and observability in production.
- Follow best practices to ensure a stable, secure, and scalable production system.

# TRY IT YOURSELF — EXERCISE 1: PROFILE PURPOSES

*Analysis exercise -- state why dev, test, and prod exist for Northstar CRM, and why a blank prod password is unacceptable.*

## STEP-BY-STEP

| STEP | WHAT TO DO |
|---|---|
| **1. Write goals** | In notes/profiles.md, write one sentence each for dev, test, and prod. |
| **2. Check reference** | Align your three sentences with the reference table below before moving on. |
| **3. Incident story** | Explain in writing why a blank prod password committed in YAML is unacceptable. |
| **4. Fixtures** | Confirm that under dev, CUS-1001 and CUS-1002 must still be callable. |

## Reference: dev / test / prod goals

```
dev   -> Local H2-friendly / verbose-safe settings
test  -> Deterministic automated tests
prod  -> Fail-fast; secrets from environment
```

## PASS CRITERIA

- Three profiles described (dev, test, prod).
- YAML-secret anti-pattern called out.
- Dev fixtures (CUS-1001 / CUS-1002) mentioned.

**KEY TAKEAWAY** TRY IT NOW -- Complete Exercise 1 before continuing: exercises/exercise-01-profile-purposes.md.

# TRY IT YOURSELF — EXERCISE 2: PROFILE YAML TODOS (STARTER)

*Hands-on starter -- fill every _____ and // TODO in three profile YAML sketches without adding real secrets.*

## STEP-BY-STEP

| STEP | WHAT TO DO |
|---|---|
| **1. Create files** | Create notes/application.yml, notes/application-dev.yml, notes/application-prod.yml. |
| **2. Fill TODOs** | Replace every _____ in the base, dev, and prod sketches using the hints on the right. |
| **3. Self-check** | Confirm the prod file has no literal password strings anywhere. |
| **4. Reflect** | Record the lab evidence correlation ID: lab26-001 (or lab-request-001). |

## Starter TODOs (replace every _____)

```
# application.yml
spring.application.name: _____
server.port: _____

# application-dev.yml
northstar.integration.api-base-url:
http://localhost:_____
logging.level.com.northstar: _____

# application-prod.yml -- no real passwords
northstar.integration.api-base-url:
${NORTHSTAR_API_BASE_URL:_____}
```

## HINTS

name = northstar-crm, port = 8080, dev logging = DEBUG, prod default placeholder = CHANGE_ME.

**KEY TAKEAWAY** TRY IT NOW (STARTER) -- Complete Exercise 2 before continuing: exercises/exercise-02-profile-yaml-todos.md.

# ENVIRONMENT VARIABLES (1 OF 2)

*External key-value pairs provided by the operating system or the deployment environment -- essential for security and portability.*

Environment variables enable applications to be configured without changing any code or configuration files.

## HOW IT WORKS

1. **1. Set Variables**
   Defined in OS, shell, Docker, Kubernetes
2. **2. Read by Spring Boot**
   Spring reads env vars at startup
3. **3. Override Properties**
   Env vars override file properties
4. **4. Applied to Application**
   Values injected into configuration

## KEY BENEFITS

- Keep sensitive information out of code and configuration files.
- Easily change values per environment (dev, test, prod); ideal for cloud, container, and CI/CD deployments.
- Improves portability, maintainability, security, and compliance.

**KEY TAKEAWAY** Environment variables have higher precedence than values in application files.

# ENVIRONMENT VARIABLES (2 OF 2)

*Naming conventions, precedence, and how to set environment variables on different platforms.*

## NAMING CONVENTIONS

| Property Name | Environment Variable |
|---|---|
| **spring.datasource.url** | SPRING_DATASOURCE_URL |
| **spring.datasource.username** | SPRING_DATASOURCE_USERNAME |
| **server.port** | SERVER_PORT |
| **northstar.integration.timeout-ms** | NORTHSTAR_INTEGRATION_TIMEOUT_MS |

Uppercase letters; dots (.) and hyphens (-) in property names are replaced with underscores (_).

## PRECEDENCE (HIGH TO LOW)

- 1. Command-line arguments
- 2. Environment variables
- 3. application-{profile}.yml / .properties
- 4. application.yml / .properties
- 5. Defaults in the code

## EXAMPLES: SETTING ENVIRONMENT VARIABLES

```
# Linux/macOS            # Windows (PowerShell)         # Docker
export SPRING_PROFILES_ACTIVE=prod   $env:SPRING_PROFILES_ACTIVE="prod"
docker run -e SPRING_PROFILES_ACTIVE=prod app
```

**KEY TAKEAWAY** Use environment variables for secrets, credentials, URLs, API keys, and environment-specific settings.

# EXTERNALIZED CONFIGURATION (1 OF 2)

*Storing application settings outside the application code so behavior can change without recompiling or redeploying.*

Spring Boot makes it easy to externalize configuration and provides a flexible way to manage settings across environments.

## HOW IT WORKS

1. **1. External Source**
   Configuration stored outside the codebase
2. **2. Loaded by Spring Boot**
   Loaded from external sources at startup
3. **3. Binds to Properties**
   Bound to environment, beans, components
4. **4. Application Runs**
   Runs with the external configuration applied

## KEY BENEFITS

- Change configuration without code changes; environment-agnostic deployments (dev, test, prod).
- Keep sensitive information out of code (security).
- Easier to manage and maintain; better portability across platforms and clouds.

**KEY TAKEAWAY** Externalized configuration enables flexibility, security, and scalability across all environments.

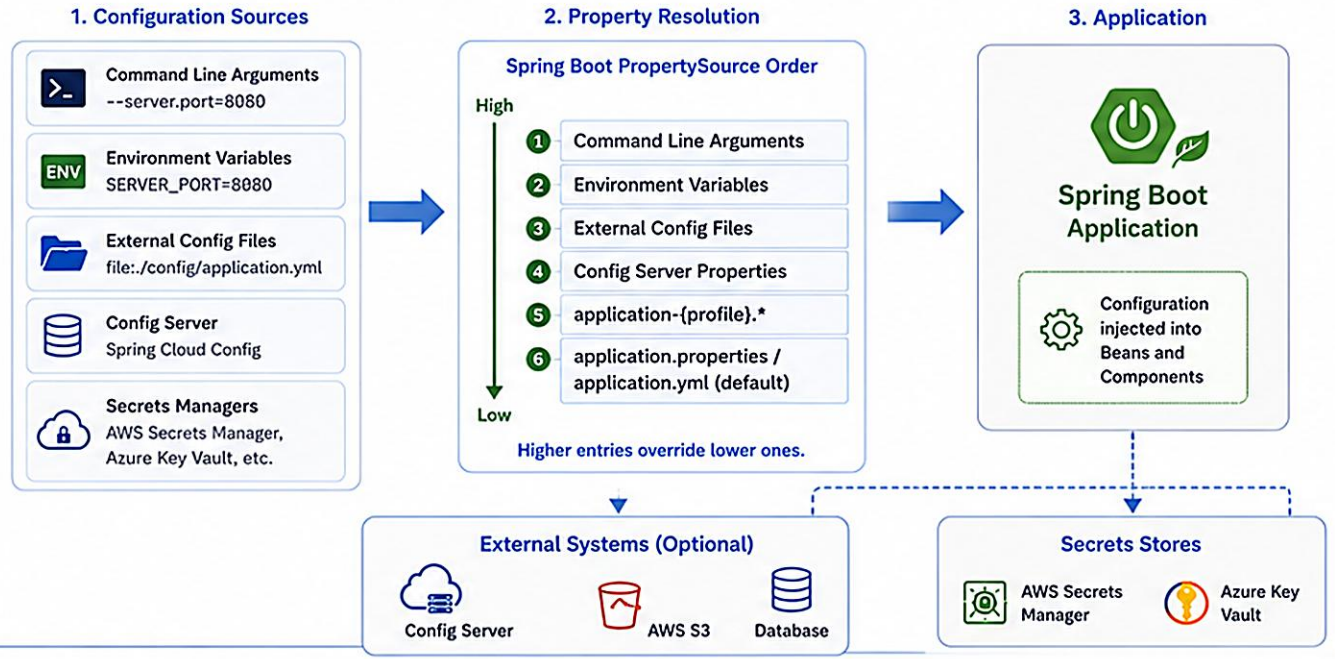
# Externalized Configuration in Spring Boot

Externalized configuration lets you store configuration outside the application, so it can be changed without rebuilding the application.

## Why Externalize?

- Change configuration without rebuilding or redeploying.
- Adapt easily across different environments.
- Keep sensitive information out of source code.
- Promote 12-Factor App principles.
- Centralize and standardize configuration management.

## How Externalized Configuration Works




## Key Benefits

- **Security** — Keep secrets and sensitive data out of code.
- **Flexibility** — Change settings per environment without code changes.
- **Scalability** — Easily manage configuration in distributed systems.
- **Portability** — Run the same artifact across different environments.
- **Maintainability** — Centralized, consistent and versioned configuration.

## Examples

### Environment Variable

```
export SERVER_PORT=8080
export DB_URL=jdbc:postgresql://db:5432/app
```

OS environment variables override file properties.

### External File (application.yml)

```
server:
  port: 8080
spring:
  datasource:
    url: jdbc:postgresql://db:5432/app
```

Place file outside the jar: fille:./config/application.yml

### Command Line Argument

```
java -jar app.jar \
  --spring.profiles.active=prod \
  --db.url=jdbc:postgresql://db:5432/app
```

Highest priority – Overrides all other sources.

## Best Practices

- Never commit secrets to source control.
- Use environment variables or secret stores for sensitive data.
- Keep defaults in the application for developer convenience.
- Document required configurations and their sources.
- Use Config Server for centralized management in microservices.

Externalized configuration is the foundation for building modular, secure and environment-agnostic applications.

Spring Boot

# EXTERNALIZED CONFIGURATION (2 OF 2)

*External source examples, a before/after code comparison, and best practices for externalizing configuration.*

## EXAMPLES OF EXTERNAL SOURCES

- Property/YAML files: application.properties / application.yml.
- Environment variables and command-line arguments.
- Config Server (Spring Cloud Config).
- Secrets management solutions (AWS Secrets Manager, Vault, etc.).

## BEST PRACTICES

**Store Config Externally** | **Env Vars for Secrets** | **12-Factor Principle** | **Document Defaults**

**Before (Hard-coded)**

```
@Value("${db.url}")
private String dbUrl = "jdbc:mysql://localhost:3306/bank";
```

**After (Externalized)**

```
@Value("${db.url}")
private String dbUrl;  // Value comes from an external source
```

**KEY TAKEAWAY** Store configuration in external sources, not in code. Follow the 12-factor app principle: strict separation of config from code.

# TRY IT YOURSELF — EXERCISE 3: CONFIGURATIONPROPERTIES SKETCH

*Documentation exercise -- sketch NorthstarIntegrationProperties fields and the prod fail-fast rule, with no real secrets.*

## STEP-BY-STEP

| STEP | WHAT TO DO |
|---|---|
| **1. Fields** | In notes/northstar-props.md, list placeholder fields: apiBaseUrl, apiKey (env-only in prod), connectTimeoutMs. |
| **2. Prefix** | Propose the YAML prefix northstar.integration for the typed binding. |
| **3. Fail-fast** | Write the rule: prod must fail startup if DB_USERNAME / DB_PASSWORD / NORTHSTAR_API_KEY are missing. |
| **4. .env.example** | List placeholder keys only -- never real values. |

## Sketch: northstar.integration.*

```
northstar:
  integration:
    api-base-url: ${NORTHSTAR_API_BASE_URL}
    api-key: ${NORTHSTAR_API_KEY}   # env-only, prod
    connect-timeout-ms: ${NORTHSTAR_TIMEOUT_MS:2000}
```

## PASS CRITERIA

- Three fields listed (apiBaseUrl, apiKey, connectTimeoutMs).
- Fail-fast prod rule written.
- No real secrets anywhere in the sketch.

**KEY TAKEAWAY** TRY IT NOW -- Complete Exercise 3 before continuing: exercises/exercise-03-config-properties-sketch.md.

# PROPERTY RESOLUTION (1 OF 2)

*Spring Boot uses a well-defined order to resolve configuration properties. When a property is defined in multiple places, the highest-precedence source wins.*

Understanding this order helps you predict which value will be used and avoid unexpected overrides.

## HOW PROPERTY RESOLUTION WORKS

1

**1. Multiple Sources**

Properties can be defined in many places

2

**2. Ordered Lookup**

Spring checks sources highest to lowest

3

**3. First Match Wins**

The first occurrence of a property is used

4

**4. Applied to Context**

The resolved value is bound to the app

## KEY TAKEAWAYS

- The highest-precedence source wins; external sources override internal files.
- Profiles and environments affect values; use this order to design predictable configuration.
- Great for troubleshooting and debugging unexpected overrides.

**KEY TAKEAWAY** When a property is found in a higher source, Spring ignores all lower sources for that property.

# Spring Boot Property Resolution

Spring Boot resolves properties from multiple sources in a fixed order. Higher priority sources override lower priority sources.

## Property Source Precedence (High → Low Priority)

HIGHEST PRIORITY

- Command Line Arguments — --server.port=9090
- Environment Variables — SERVER_PORT=9090
- Java System Properties — -Dserver.port=9090
- JNDI Attributes — java:comp/env/...
- ServletContext Init Parameters — web.xml or programmatic
- ServletConfig Init Parameters — web.xml or programmatic
- application-{profile}.yml/properties — (profile-specific)
- application.yml/properties — (default)
- External Config (file:, classpath:/config/, optional: )
- Config Server — Spring Cloud Config
- Default Properties — @PropertySource, defaults
- Spring Boot Defaults — (lowest priority)

LOWEST PRIORITY

## How Property Resolution Works

1. **Collect** — Spring Boot collects all available property sources.
2. **Order** — Sources are ordered by precedence (high → low).
3. **Resolve** — Each property is resolved from the highest priority source that contains it.
4. **Use** — Resolved properties are injected into beans and components.

### Example: Resolving server.port

| Priority | Property Source | server.port | Result |
|---|---|---|---|
| ① 1 | Command Line Arguments | `--server.port=9090` | 9090 (Resolved) |
| ② 2 | Environment Variables | `SERVER_PORT=8080` | (Ignored) |
| ③ 3 | application-prod.yml | `server.port: 7070` | (Ignored) |
| ⑧ 8 | application.yml (default) | `server.port: 6060` | (Ignored) |
| ⑫ 12 | Spring Boot Default | `server.port: 8080` | (Ignored) |

✓ The value from the highest priority source that provides the property is used.

## Key Behavior

**Highest priority wins**
Once a property is found at a higher level, lower levels are ignored.

**Profile-specific overrides default**
application-{profile}.yml overrides application.yml.

**Same level?**
Later sources override earlier ones at the same level.

**Not found anywhere?**
Spring Boot uses its default value (if available).

## Effective Properties Example

```
--server.port=9090      # from Command Line
SERVER_PORT=8080        # from Environment Variable
server.port=7070        # from application-prod.yml
server.port=6060        # from application.yml
server.port=8080        # Spring Boot Default
```

`Effective server.port = 9090` ✓

## Property Usage in Code

### @Value Injection

```
@Value("${app.name}")
privete String appName;
```

### @ConfigurationProperties

```
@ConfigurationProperties(prefix = "app")
public class AppProperties {
    private String name;
}
```

### Environment API

```
@Autowired
privete Environment env;

env.getProperty("app.name");
```

### Relaxed Binding

app.name, APP_NAME, app-name, app_name all map to the same property.

## Best Practices

- Keep sensitive data in environment variables or secrets management tools.
- Use profiles to separate configuration for different environments.
- Document important properties with default values.
- Externalize configuration to avoid code changes across environments.

**Tip:** Use Spring profiles, environment variables, and external configuration for clean, secure, and flexible configuration management.

Spring Boot

# PROPERTY RESOLUTION (2 OF 2)

*The complete property source precedence order, and a worked example of the same property defined in five different places.*

## PROPERTY SOURCE PRECEDENCE (HIGHEST TO LOWEST)

- 1. Command-line arguments (e.g. --app.name=MyApp-CLI)
- 2. Environment variables (e.g. APP_NAME=MyApp-ENV)
- 3. Java system properties (e.g. -Dapp.name=MyApp-SYS)
- 4. JNDI attributes from java:comp/env
- 5. ServletConfig / ServletContext init parameters
- 6. application-{profile}.yml / .properties (profile-specific files)
- 7. application.yml / .properties (base file)
- 8. @PropertySource files
- 9. Default properties (via setDefaultProperties) -- lowest precedence

## EXAMPLE: SAME PROPERTY, MULTIPLE PLACES

| Source (High to Low) | app.name Value |
|---|---|
| **Command-line argument** | MyApp-CLI |
| **Environment variable** | MyApp-ENV |
| **application-prod.yml** | MyApp-ProdYML |
| **application.yml** | MyApp-YML |
| **Default in @Value / code** | MyApp-Default |

If app.name is passed as a command-line argument, 'MyApp-CLI' wins and every lower source is ignored for that key.

**KEY TAKEAWAY** Prefer external configuration for flexibility and security; test with different sources to ensure correct behavior.

# TRY IT YOURSELF — EXERCISE 4: PROPERTY OVERRIDE ORDER

*Architecture exercise -- rank CLI, env, profile YAML, and base YAML by precedence, then write both activation styles.*

## STEP-BY-STEP

| STEP | WHAT TO DO |
|---|---|
| **1. Rank** | In notes/override-order.md, number the four sources highest to lowest. |
| **2. Check reference** | Compare your ranking to the reference table on the right; correct mistakes. |
| **3. Activation pair** | Write example activations: -Dspring.profiles.active=dev and SPRING_PROFILES_ACTIVE=prod. |
| **4. Measurement plan** | Note that Lab 26 asks for measured override evidence -- defer capturing it to the lab. |

## Reference: precedence (highest first)

```
1. Command-line args / -D
2. Environment variables
3. Profile-specific YAML
4. Base application.yml
```

## PASS CRITERIA

- Order matches the reference table.
- Both activation styles listed.
- Lab measurement deferred explicitly.

**KEY TAKEAWAY** TRY IT NOW -- Complete Exercise 4 before continuing: exercises/exercise-04-override-order.md.

# MANAGING SENSITIVE INFORMATION (1 OF 2)

*Passwords, API keys, tokens, and database credentials must be protected -- never hard-coded in the application.*

Spring Boot provides multiple ways to externalize and secure sensitive configuration so it is never hard-coded in the application.

## WHY IT MATTERS

### Protects Access

Protects against unauthorized access and data breaches.

### Compliance

Helps meet security and compliance requirements.

### Safe Sharing

Enables safe sharing and deployment across environments.

### Reduced Exposure

Reduces risk of accidental exposure in code or logs.

## KEY TAKEAWAYS

- Never store secrets in source code or configuration files in version control.
- Use externalized configuration and secret management tools; limit access to secrets and rotate them regularly.
- Audit usage and avoid logging sensitive information.

**KEY TAKEAWAY** Never store secrets in source code or configuration files in version control.

# MANAGING SENSITIVE INFORMATION (2 OF 2)

*Concrete ways to manage sensitive information, best practices, and a worked application.yml example.*

## WAYS TO MANAGE SENSITIVE INFORMATION

- **Environment Variables —** set secrets as environment variables; Spring Boot reads them at runtime.
- **Config Server + Vault —** store secrets in HashiCorp Vault and access via Spring Cloud Config.
- **Cloud Secret Managers —** AWS Secrets Manager, Azure Key Vault, Google Secret Manager.
- **External Files (Not in Repo) —** store secrets in external files mounted at runtime, never in the repo.
- **Encrypted Properties —** encrypt values and decrypt at runtime using Jasypt or similar libraries.

### Example: application.yml

```
spring:
  datasource:
    username: ${DB_USERNAME}
    password: ${DB_PASSWORD}  # from environment
```

## BEST PRACTICES

- Externalize all secrets.
- Never commit .env (.gitignore it).
- Grant least-privilege access.
- Rotate secrets regularly.
- Never log secret values.

## GITIGNORE + .ENV.EXAMPLE PATTERN

### .env.example (committed, placeholders only)

```
DB_USERNAME=changeme
DB_PASSWORD=changeme
NORTHSTAR_API_KEY=changeme

# .env itself is .gitignore'd -- never commit it
```

**KEY TAKEAWAY** Treat secrets like keys to your house -- store them securely, control access, and change them regularly.

# SECRETS MANAGEMENT BASICS (1 OF 2)

*Secrets are sensitive pieces of information such as passwords, API keys, tokens, and certificates -- managing them securely is critical.*

A secrets manager provides a secure, centralized place to store secrets and control access to them.

## WHY USE A SECRETS MANAGER?

### Centralized Storage

One secure, centralized place to store secrets.

### Fine-grained Access

Fine-grained access control per secret and consumer.

### Automatic Rotation

Automatic secret rotation without code changes.

### Audit Logging

Audit logging and monitoring of secret access.

## COMMON SECRETS

- Database Passwords
- API Keys & Tokens
- Email Credentials
- Encryption Keys/Certs
- Third-Party Credentials

**KEY TAKEAWAY** A secrets manager provides a secure, centralized place to store secrets and controls access to them.

# SECRETS MANAGEMENT BASICS (2 OF 2)

*How a secrets manager works end to end, popular secrets managers, and a worked Spring Boot example.*

## HOW IT WORKS

1. **1. Store Secret**
   Secrets stored encrypted in the manager
2. **2. Authenticate**
   App authenticates using its identity
3. **3. Retrieve Secret**
   App requests the secret at runtime
4. **4. Use and Rotate**
   Secret is used and can rotate without code changes

## POPULAR SECRETS MANAGERS

| Manager | Description |
|---|---|
| **AWS Secrets Manager** | Fully managed, integrates with IAM and AWS services. |
| **Azure Key Vault** | Secure store for secrets, keys, and certificates in Azure. |
| **Google Secret Manager** | Managed service to store, manage, and access secrets. |
| **HashiCorp Vault** | Open-source and enterprise solution for secrets and encryption. |

## Example: Spring Boot + AWS Secrets Manager

```
spring:
  datasource:
    username: ${DB_USERNAME}
    password: ${DB_PASSWORD}
aws:
  secretsmanager:
    secret-name: myapp/prod/db-credentials
    region: us-east-1
```

**KEY TAKEAWAY** Rotate secrets regularly, grant least-privilege access, and never log or expose secret values.

# CONFIGURATION ACROSS ENVIRONMENTS (1 OF 2)

*Different environments have different requirements. Spring Profiles and externalized configuration let us manage them cleanly and safely.*

The goal is to promote consistency, reduce risk, and make deployments reliable -- keep code identical, change only the configuration.

## THE BIG PICTURE

1. **Development**
   Local machines, fast iteration
2. **Test**
   Automated tests, isolated data
3. **Staging**
   Pre-production, UAT/validation
4. **Production**
   Live users, performance tuned

## KEY TAKEAWAYS

- Keep code identical; change configuration, not code.
- Use profiles to separate environment concerns; store sensitive information securely.
- Externalize configuration for portability and scalability; test in lower environments before promoting to production.

**KEY TAKEAWAY** Use Spring Profiles and externalized configuration to keep code the same and settings environment-specific.

# Configuration Across Environments

Use Spring Profiles and externalized configuration to manage environment-specific settings consistently across Development, Test, and Production.

**Goal**
Same application, different configuration for each environment.

## 1. DEVELOPMENT (dev)

For local development

**Purpose**
Fast development, debugging, local tools and mocks

**Active Profile** `dev`

**Key Characteristics**
- Debug logging
- Local database (H2 / MySQL)
- External services mocked
- Relaxed security
- Developer conveniences enabled

**application-dev.yml**

```yaml
server:
  port: 8080
logging:
  level:
    root: DEBUG
datasource:
  url: jdbc:h2:mem:devdb
feature:
  beta-enabled: true
```

## 2. TEST (test)

For automated & manual testing

**Purpose**
Validate functionality, integrations and performance

**Active Profile** `test`

**Key Characteristics**
- Info logging
- Test database (isolated)
- Real or mock integrations
- Test security setup
- Seeded test data

**application-test.yml**

```yaml
server:
  port: 8081
logging:
  level:
    root: INFO
datasource:
  url: jdbc:postgresql://test-db:5432/app_test
feature:
  beta-enabled: false
```

## 3. PRODUCTION (prod)

For live users and operations

**Purpose**
Stability, security, performance and reliability

**Active Profile** `prod`

**Key Characteristics**
- Warn / Error logging
- Production database (HA)
- Real external services
- Strict security
- Performance optimizations

**application-prod.yml**

```yaml
server:
  port: 80
logging:
  level:
    root: WARN
datasource:
  url: jdbc:postgresql://prod-db:5432/app
feature:
  beta-enabled: false
```

## How It Works

1. **Package Once**
   Build the application artifact once (JAR/WAR/Docker image).
2. **Externalize Configuration**
   Environment-specific properties are externalized.
3. **Activate Profile**
   Activate the appropriate profile at runtime for each environment.
4. **Run with Environment Settings**
   The application runs using the correct configuration for that environment.

## Example: Property Differences

| Property | Development (dev) | Test (test) | Production (prod) | Description |
|---|---|---|---|---|
| server.port | 8080 | 8081 | 80 | Port exposed by the application |
| logging.level.root | DEBUG | INFO | WARN | Logging level |
| datasource.url | jdbc:h2:mem:devdb | jdbc:postgresql://test-db:5432/app_test | jdbc:postgresql://prod-db:5432/app | Database connection URL |
| feature.beta-enabled | true | false | false | Feature toggle example |

## Ways to Activate Profiles

**Command Line**

```
--spring.profiles.active=dev

java -jar app.jar \
--spring.profiles.active=prod
```

**Environment Variable**

```
SPRING_PROFILES_ACTIVE=dev

export SPRING_PROFILES_ACTIVE=prod
```

**application.properties / application.yml**

```
spring.profiles.active=dev
```

**Config Server / Orchestrator**
Set profile via Config Server, Kubernetes, Docker, etc.

**Best Practice:** Keep configuration external, use profiles for differences, and keep secrets out of source code.

Secure · Consistent · Maintainable · Scalable

# CONFIGURATION ACROSS ENVIRONMENTS (2 OF 2)

*A worked database configuration example across all four environments, file naming conventions, and how to activate a profile.*

## EXAMPLE: DATABASE CONFIGURATION

| Property | development | test | staging | production |
|---|---|---|---|---|
| **spring.datasource.url** | localhost:5432/bank | test-db:5432/bank | staging-db:5432/bank | prod-db.cluster:5432/bank |
| **spring.datasource.username** | dev_user | test_user | staging_user | prod_user |
| **spring.datasource.password** | dev_pass | test_pass | staging_pass | (from secret store) |
| **logging.level.root** | DEBUG | INFO | INFO | WARN |

## PROFILE-SPECIFIC FILE NAMING

- application.yml -- default (all environments)
- application-dev.yml -- Development
- application-test.yml -- Test
- application-staging.yml -- Staging
- application-prod.yml -- Production

## WAYS TO ACTIVATE A PROFILE

**Command Line** **Environment Variable** **application.yml**

**IDE Run Configuration**

**KEY TAKEAWAY** Right configuration in the right environment leads to secure, reliable, and scalable applications.

# TRY IT YOURSELF — EXERCISE 5: ACTIVATION COMMAND DRILL

*Analysis exercise -- write Windows and macOS profile activation commands for your notes.*

## STEP-BY-STEP

| STEP | WHAT TO DO |
|---|---|
| **1. -D form** | In notes/activation-commands.md, write the -Dspring.profiles.active=dev style (or the mvn equivalent). |
| **2. Env form** | Write both the Windows PowerShell and macOS/Linux environment-variable forms. |
| **3. Prod caution** | Note: never run the prod profile without required env vars set -- expect fail-fast. |
| **4. Boundary** | Do not start the full Lab 26 app in this exercise unless the instructor asks. |

## Windows vs macOS activation

```
# -D form (via Maven)
mvn spring-boot:run -Dspring-boot.run.arguments=
  --spring.profiles.active=dev

# Windows PowerShell
$env:SPRING_PROFILES_ACTIVE="test"

# macOS / Linux
export SPRING_PROFILES_ACTIVE=test
```

## PASS CRITERIA

- -D example present.
- Env-var example present (both OS).
- Prod fail-fast caution written.

**KEY TAKEAWAY** TRY IT NOW -- Complete Exercise 5 before continuing: exercises/exercise-05-activation-drill.md.

# CONFIGURATION BEST PRACTICES (1 OF 2)

*Following best practices ensures your configuration is secure, maintainable, portable, and easy to manage across different environments.*

## FOLLOW THESE CORE PRINCIPLES

**Externalize**

Keep configuration outside the code.

**Organize**

Use a clear structure and naming conventions.

**Secure**

Protect secrets and sensitive data.

**Environment-Aware**

Use profiles and variables for environment-specific config.

**Consistent**

Maintain consistency across all environments and services.

## BEST PRACTICES FOR SPRING CONFIGURATION (1-5)

| # | Practice | What It Means |
|---|---|---|
| **1** | Externalize Everything | Move all configurable values out of the code. |
| **2** | Use the Right Source | Prefer environment variables, Config Server, or Vault for sensitive values. |
| **3** | Follow Naming Conventions | Use meaningful property names (e.g. app.datasource.url). |
| **4** | Use Profiles Wisely | Keep common settings in application.yml and override only what's needed. |
| **5** | Separate Config by Concern | Group properties by feature or module. |

**KEY TAKEAWAY** Good configuration is the foundation of a resilient, secure, and scalable application.

# CONFIGURATION BEST PRACTICES (2 OF 2)

*Practices 6-10, what to avoid, and an example of a good configuration file structure.*

## BEST PRACTICES FOR SPRING CONFIGURATION (6-10)

- **6. Never Commit Secrets —** use .gitignore and secret management tools.
- **7. Use Placeholders and Defaults —** provide sensible defaults, but never for secrets; use ${...:default}.
- **8. Validate Configuration —** fail fast for missing or invalid critical properties.
- **9. Document Configuration —** maintain clear documentation for required properties and their meaning.
- **10. Monitor and Audit —** track changes to configuration and access to secrets.

## WHAT TO AVOID

**Hard-coding Values**

**Plain-text Secrets**

**One Config for All Envs**

**Skipping Validation**

## Example: Good Config Structure

```
src/main/resources/
├── application.yml           # common
├── application-dev.yml       # development
├── application-test.yml      # test
├── application-staging.yml   # staging
├── application-prod.yml      # production
└── application-secrets.yml   # local, .gitignored

# Keep common properties in application.yml
# and override only what changes per profile.
```

**KEY TAKEAWAY** Externalize everything, never commit secrets, and validate configuration at startup -- fail fast for missing critical properties.

# TRY IT YOURSELF — EXERCISE 6: LAB 26 READINESS (CAPSTONE)

*Documentation capstone -- ready the lab26-crm path and secret hygiene before starting the graded Lab 26.*

## STEP-BY-STEP

| STEP | WHAT TO DO |
|---|---|
| **1. Depends on** | In notes/lab26-readiness.md, note the Lab 25 layered CRM dependency. |
| **2. Path** | Write the project path: examples/lab26-crm. |
| **3. Hygiene** | Confirm .gitignore will cover .env and target/. |
| **4. Defer** | Note that transactions (Lab 27) and security (Lab 28) wait -- do not build them yet. |

## .gitignore hygiene check

```
# Must be covered before Lab 26 starts
.env
target/

# Committed instead (placeholders only)
.env.example
```

## PASS CRITERIA

- Lab 25 dependency noted.
- Path written correctly (examples/lab26-crm).
- .env / target/ hygiene stated.

**KEY TAKEAWAY** TRY IT NOW -- Complete Exercise 6 (capstone) before Lab 26: exercises/exercise-06-lab26-readiness.md.

# LAB OVERVIEW -- SPRING PROFILES AND CONFIGURATION

*Lab 26: Spring Profiles and Configuration -- Northstar CRM Environments*

## BUSINESS SCENARIO

- Incidents from dev settings leaking into production (open H2 console, blank DB password, verbose SQL) are unacceptable.
- Leadership freezes: running configuration depends on where the app is deployed; production credentials arrive only via environment variables.
- Missing required prod properties must fail fast at startup -- never connect with blank passwords.

## LEARNING OBJECTIVES

- Explain Spring's property-source override order: CLI > env > application-{profile}.yml > application.yml > code defaults.
- Convert application.properties to application.yml and structure dev/test/prod profile files.
- Activate a profile with -Dspring.profiles.active and with SPRING_PROFILES_ACTIVE.
- Bind externalized configuration with @ConfigurationProperties instead of scattered @Value.
- Apply secrets-handling practices while keeping CRM fixtures working under dev.

## WHAT YOU BUILD

- Copy lab25-crm to lab26-crm; convert to application.yml with dev/test/prod profile files; activate profiles two ways; prove CLI > env > profile-YAML override order with northstar.integration.timeout-ms; add a validated NorthstarIntegrationProperties; ship .env.example with placeholders only; confirm dual green mvn test under test and a CUS-1001 smoke test under dev.

**KEY TAKEAWAY** Fixtures CUS-1001 (Amina Khan, ACTIVE) and CUS-1002 (Ravi Singh, PROSPECT) must still work under dev; correlation ID lab26-001 anchors every verification.

# LAB TASKS AND DELIVERABLES

*Northstar CRM Environments -- implementation steps, deliverables, and the evaluation rubric.*

| # | Implementation Step |
|---|---|
| 1 | Branch lab25-crm to lab26-crm; inventory existing properties before deleting anything. |
| 2 | Convert shared defaults to application.yml (name, port, management, logging). |
| 3 | Author application-dev.yml and application-test.yml (H2, logging levels, mocks). |
| 4 | Author application-prod.yml with env-only secrets -- no default passwords. |
| 5 | Activate profiles two ways: -Dspring-boot.run.profiles and SPRING_PROFILES_ACTIVE. |
| 6 | Prove override order with northstar.integration.timeout-ms (YAML -> env -> CLI). |
| 7 | Bind NorthstarIntegrationProperties (@Validated) and ship .env.example. |
| 8 | Tests under test profile + CUS-1001/CUS-1002 smoke under dev; failure experiments + secrets hygiene. |

## EXPECTED DELIVERABLES

- application.yml + dev/test/prod profile files.
- NorthstarIntegrationProperties + enable config.
- .env.example placeholders only.
- Evidence of -D and env profile activation; fail-fast prod startup evidence.
- Override-order notes with measurements; dual green tests under test; no secrets or target/ committed.

## RUBRIC (100 MARKS)

- Environment/Structure 10 · Core Impl 30 · Activation/Override 15 · Externalized/Fail-fast 15 · Failure Handling 10 · Security/Secrets 10 · Docs 10

**KEY TAKEAWAY** Real secrets in Git are an honor violation; prod YAML with default passwords loses security/fail-fast marks; no override evidence loses activation marks.

# MODULE 26 SUMMARY

*In this module, we explored how to build flexible, secure, and environment-aware Spring Boot applications using Spring Profiles and externalized configuration.*

## KEY CONCEPTS COVERED

### Properties & YAML

application.properties and application.yml, and when to use each.

### Spring Profiles

Define, activate, and structure environment-specific configuration.

### Externalized Config

Environment variables, CLI args, and property source precedence.

### Property Resolution

The full precedence order Spring uses to resolve conflicting values.

### Secrets Management

Never hard-code secrets; environment variables and secrets managers.

### Best Practices

Externalize, organize, secure, validate, and fail fast.

## MODULE FLOW RECAP

1. **YAML & Properties**
2. **Spring Profiles**
3. **Externalize & Resolve**
4. **Secure Secrets**
5. **Lab: Northstar CRM**

**KEY TAKEAWAY** Good configuration leads to flexible applications. Secure configuration leads to trustworthy applications.

# KNOWLEDGE CHECK (1 OF 3)

*Section 1: Multiple Choice -- test your understanding of Spring Profiles, configuration files, and profile activation.*

**1. What is the primary purpose of Spring Profiles?**

**A.** To encrypt configuration files

**B.** To create database backups

**C. To provide environment-specific configuration**

**D.** To define user roles and permissions

**2. Which file is the default configuration file in a Spring Boot application?**

**A.** config.properties

**B. application.properties**

**C.** spring-config.yml

**D.** bootstrap.properties

**3. Which of the following is a valid way to activate a profile?**

**A.** spring.profiles.active=dev (in application.properties)

**B.** --spring.profile=dev

**C.** -Dspring.profile.active=dev

**D. All of the above**

**4. Which of the following has the highest precedence in property resolution?**

**A.** application.properties

**B. Environment variables**

**C.** application-{profile}.yml

**D.** Default values in code

**KEY TAKEAWAY** Spring Profiles provide environment-specific configuration; environment variables outrank profile and base files in property resolution.

# KNOWLEDGE CHECK (2 OF 3)

*Two more multiple-choice questions, plus Section 2: Match the Following.*

**5. How can you externalize configuration in a Spring Boot application?**

**A.** Store values in code constants

**B. Use application.yml and environment variables**

**C.** Rebuild the application for every change

**D.** Hardcode values in beans

**6. Which of the following is NOT a recommended practice for configuration?**

**A.** Externalize properties

**B. Store secrets in plain text in code**

**C.** Use profiles for environment differences

**D.** Validate configuration at startup

## SECTION 2: MATCH THE FOLLOWING

| Concept | Description |
|---|---|
| **1. Spring Profile** | b. Defines a set of properties for a specific environment. |
| **2. Externalized Configuration** | e. Keeping configuration outside of application code. |
| **3. Environment Variable** | c. External source used to override application properties. |
| **4. Property Resolution** | d. Process of Spring finding the final value of a property. |
| **5. Secrets Manager** | a. Used to store and retrieve sensitive information securely. |

**KEY TAKEAWAY** A property must be moved out of code (externalized) before profiles, environment variables, or a secrets manager can manage it.

# KNOWLEDGE CHECK (3 OF 3)

*Section 3: True or False, Section 4: Scenario-Based, and Section 5: Short Answer -- with model answers.*

## SECTION 3: TRUE OR FALSE

| Statement | Answer |
|---|---|
| **Spring Profiles allow running the same app with different configs.** | True |
| **application-{profile}.properties overrides application.properties when active.** | True |
| **Environment variables cannot override properties defined in files.** | False |
| **Storing database passwords in application.properties is a best practice.** | False |
| **The order of property resolution determines the final property value.** | True |

## SECTION 4: SCENARIO-BASED

- **1. Different DB URLs per environment —** define shared defaults in application.yml, then override spring.datasource.url in application-dev/test/prod.yml, and activate the right profile per deployment.
- **2. An API key that must not live in code or files —** store it as an environment variable (or in a secrets manager), reference it as ${API_KEY} with no default, and let startup fail fast if it's missing.

## SECTION 5: SHORT ANSWER

- **1. Three best practices —** externalize configuration; never commit secrets (use env vars/secrets managers); use profiles to separate dev/test/prod settings.
- **2. Benefits of externalized configuration —** no code changes needed per environment; secrets stay out of source control; easier portability and scalability.

**KEY TAKEAWAY** Review these concepts and apply them in the lab -- strong configuration leads to secure, maintainable, environment-ready applications.

*"Configure once. Run anywhere."*

MODULE 26 COMPLETE

# Spring Configuration, Profiles and Environments

You can now build Spring Boot applications that adapt safely across development, test, and production using profiles, externalized configuration, and secure secrets handling.